

Thermal Reserve

Coordinated thermostat setbacks that keep Southcentral Alaska's gas demand under supply on the coldest days of a multi-day cold snap.

Join from your phone: thermal-reserve.vercel.app/home



THE PROBLEM

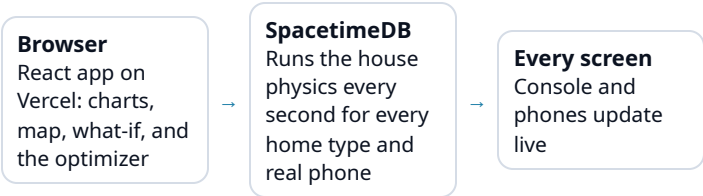
- Enstar's president warned lawmakers the utility could enter this winter about **3 Bcf short** of gas, the equivalent of **18 days without heat** in sustained cold.¹
- On Aug 7, 2026, Cook Inlet storage held about **6.54 Bcf** of its 13 Bcf working capacity.¹
- In a shortage, Enstar would ask **large commercial and industrial customers to cut back first**, before homes, and imported LNG may not arrive until **late 2029**.²
- Thermostat programs elsewhere save gas during short events, but homes reheat afterward ("snapback"). In ConEd's pilot, snapback erased **52%** of the savings.³

WHAT WE BUILT

A **simulation**, not control of real thermostats. An operator console plans setbacks for a simulated fleet of Anchorage homes (24 home types, shown as 1,000 dots on a map). An optimizer picks the smallest total temperature drop that keeps each day's gas demand under supply, planning across the whole cold snap so reheating lands on days with spare gas.

Households join from a phone, see their indoor temperature and gas saved, and can opt out with one tap. The server never lets a home drop below **62°F** (60°F at the lowest setting), and homes that need steady heat are exempt.

HOW IT WORKS



The browser and the server run the same physics file, and a test checks that they agree.

SOURCES (EVERY LINK OPENED AND CHECKED ON OCT 3, 2026)

1. Anchorage Daily News editorial board, "The winter nobody wants to imagine," Aug 8, 2026. adn.com/opinions/editorials/2026/08/08/editorial-the-winter-nobody-wants-to-imagine

2. Homer News, "Winter gas shortfall comes into sharper focus for Southcentral Alaska," Sep 10, 2026. homernews.com/2026/09/10/winter-gas-shortfall-comes-into-sharper-focus-for-southcentral-alaska

3. Con Edison, "Gas Demand Response Report on Pilot Performance – 2019/2020," Jul 1, 2020 (NY PSC Case 17-G-0606). [documents.dps.ny.gov \(DocRefId 7FA119BD-B04D-4378-95A3-BB26697815AC\)](https://documents.dps.ny.gov/publicDocIndex?docID=7FA119BD-B04D-4378-95A3-BB26697815AC)

4. Anchorage Daily News, "Results are in for latest Southcentral Alaska Energy Watch drill," Nov 6, 2012. adn.com/energy/article/results-are-latest-southcentral-alaska-energy-watch-drill/2012/11/06

5. ACIS Web Services, Regional Climate Centers: daily data for Anchorage Ted Stevens International Airport, 1996–2025. rcc-acis.org/docs_webservices.html

6. U.S. Energy Information Administration, "Heat Content of Natural Gas Delivered to Consumers," Alaska, 2025. eia.gov/dnav/ng/ng_cons_heat_a_epg0_vgth_btucf_a.htm

7. Petroleum News, "New Enstar supplies," Mar 6, 2016. petroleumnews.com/pntruncate/978589605.shtml

8. CALMAC SCG0224, "2018-2019 Winter Load Impact Evaluation of SoCalGas Smart Therm Program," Oct 24, 2019. calmac.org/publications/SoCalGas_2019_DR_Evaluation_Report_-_PUBLIC_FINAL.pdf